ALEX LUANGXAY

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EDUCATION

Concordia University Expected Dec 2027
Bachelor of Software Engineering (CO-OP)

WORK EXPERIENCE

Caisse de dépôt et placement du Québec (CDPQ)| Software *& Data Engineer Intern, External Data Optimization Team* April 2026 - Present
(Power BI, SQL, Snowflake, Databricks, Outlook API, REST APIs, Power Automate)
Montreal, QC

- Architected and implemented a production-ready workflow that processes operational emails from a shared Outlook mailbox and converts unstructured communications into structured business requests.
- Designed the complete Snowflake data model, including email, request, relationship, history, and audit tables, with idempotent processing and transaction-safe operations.
- Developed a hierarchical retrieval engine combining conversation threading, business identifier matching, and keyword relevance scoring to provide contextual information for LLM reasoning.
- Integrated Databricks serving endpoints through REST APIs to perform multi-step AI analysis, including request classification, metadata extraction, contextual reasoning, and business action recommendations.
- Built automated decision workflows capable of creating, updating, linking, and tracking operational requests while preserving human validation checkpoints and audit history.
- Led development of executive Power BI reporting solutions, delivering strategic insights to VP-level stakeholders.
- Owned the end-to-end reporting lifecycle, from SQL development and data modeling to dashboard design, security, and maintenance.
- Built reporting visibility into $100M+ in budget allocation and expenditures across software applications, technology services, and data platforms.
- Served as sole technical owner of the reporting platform, ensuring data accuracy, reliability, security, and performance.

Caisse de dépôt et placement du Québec (CDPQ)| *Data Developer, External Market Data Team* Sept 2025 - April 2026
(Apache NiFi, Snowflake, AWS S3, Python, SQL)
Montreal, QC

- Designed, implemented, and maintained Apache NiFi data integration pipelines, developing new workflows and optimizing existing data flows.
- Orchestrated automated data ingestion using Amazon S3 and Snowflake (Stages, Tasks, and Snowpipe) to load supplier data into enterprise data warehouses.
- Developed and maintained Snowflake schemas, views, and automated workflows to improve platform performance, data reliability, and operational efficiency.
- Optimized data ingestion and transformation pipelines to support scalable analytics, reporting, and business intelligence workloads.
- Partnered with business and technical stakeholders to deliver accurate, scalable, and high-quality data solutions.

Caisse de dépôt et placement du Québec (CDPQ)| *Data & BI Engineer Intern/Student, Reference Market Data Team* Sept 2024 - Sept 2025
(Power BI, SQL Server, Snowflake)
Montreal, QC

- Developed and deployed Power BI dashboards providing visibility into enterprise cloud data migration progress, pipeline status, and key operational metrics.
- Built automation tools deployed on corporate infrastructure to improve SSIS job recovery, reduce manual intervention, and increase process efficiency.
- Implemented Delta validation and monitoring controls to detect data inconsistencies across environments and alert teams to processing issues.
- Developed, optimized, and maintained Snowflake database structures, including schemas and analytical views, to support reporting, data quality, and BI initiatives.
- Contributed to the reliability and scalability of the organization's data ecosystem through automation, monitoring, and process optimization.

PROJECT

UdeMHacks Hackathon (1st place overall & 1st best AI used) | March 2025
Built a full-stack AI-powered educational platform for medical diagnostic training, featuring dynamic clinical case generation from symptom datasets.
Developed chatbot workflows, diagnostic evaluation logic, and frontend-backend integrations in a 4-person team.

McHack Hackathon | January 2024
Developed a React and Python web application for invoice management and automated expense categorization.
Integrated AI APIs to analyze and classify user spending patterns in a 4-person team.

ConUHacks Hackathon | January 2024
Developed frontend features for a web application using React and Node.js in a 4-person team.

CodeJam Hackathon | January 2024
Built a Python-based video enhancement solution and implemented an AI-driven numerical recognition system.

TECHNICAL SKILLS

Languages: Java, C# • C++ • JavaScript/TypeScript • Python • HTML/CSS • SQL • Go (Golang)
Full-stack: Spring Boot • .NET 8 • Node.js/Express • React • Angular • Flask • React Native
Data Platforms & Analytics: PostgreSQL • MongoDB • Snowflake • SQL Server Management Studio • DBeaver • AWS S3 • Power BI • Apache NiFi • SnapLogic • Databricks • Power Automate • SwiftUI
DevOps/Infra: Docker, GitHub Actions
Testing: JUnit
Tools: Git • IntelliJ • VS Code • Postman • JIRA
Awards: UdeMHacks Hackathon (1st Place Overall & Best Use of AI)
Interests: Stock Market • Chess • Rubik's cube • Guitar • Piano • Dragon Boat Racing • Badminton • Distance running/swimming/biking (triathlon) • Snowboarding